

SPP1 / Osteopontin-CD44 Axis

An onco-immunology drug-target program: evidence, feasibility,
novel-target strategy, development plan & spatially-rationalized bispecifics

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Executive summary

Scope

23 recent onco-immunology targets scored for genetic/functional-genomic evidence, clinical feasibility, and novel-target opportunity. Deep-dive on the SPP1(osteopontin)–CD44 immunosuppressive axis with a concrete therapeutic development plan and bispecific designs.

Lead thesis

SPP1+ tumor-associated macrophages sit adjacent to exhausted CD8 T cells and drive immune escape via SPP1→CD44 (IO-validated, 2025). A NAKED function-blocking anti-CD44 antibody that occupies the SPP1-binding region — not a cytotoxic payload, not an HA-blocker — is the lead modality.

Why receptor-side

Every SPP1-receptor interface has been drugged, but no agent has succeeded on this axis in oncology: integrin trials were negative (cilengitide CENTRIC, volociximab, POSEIDON) and the CD44 ADC (bivatuzumab-mertansine) was fatally toxic. Two guardrails engineer around those failures.

Novel whitespace

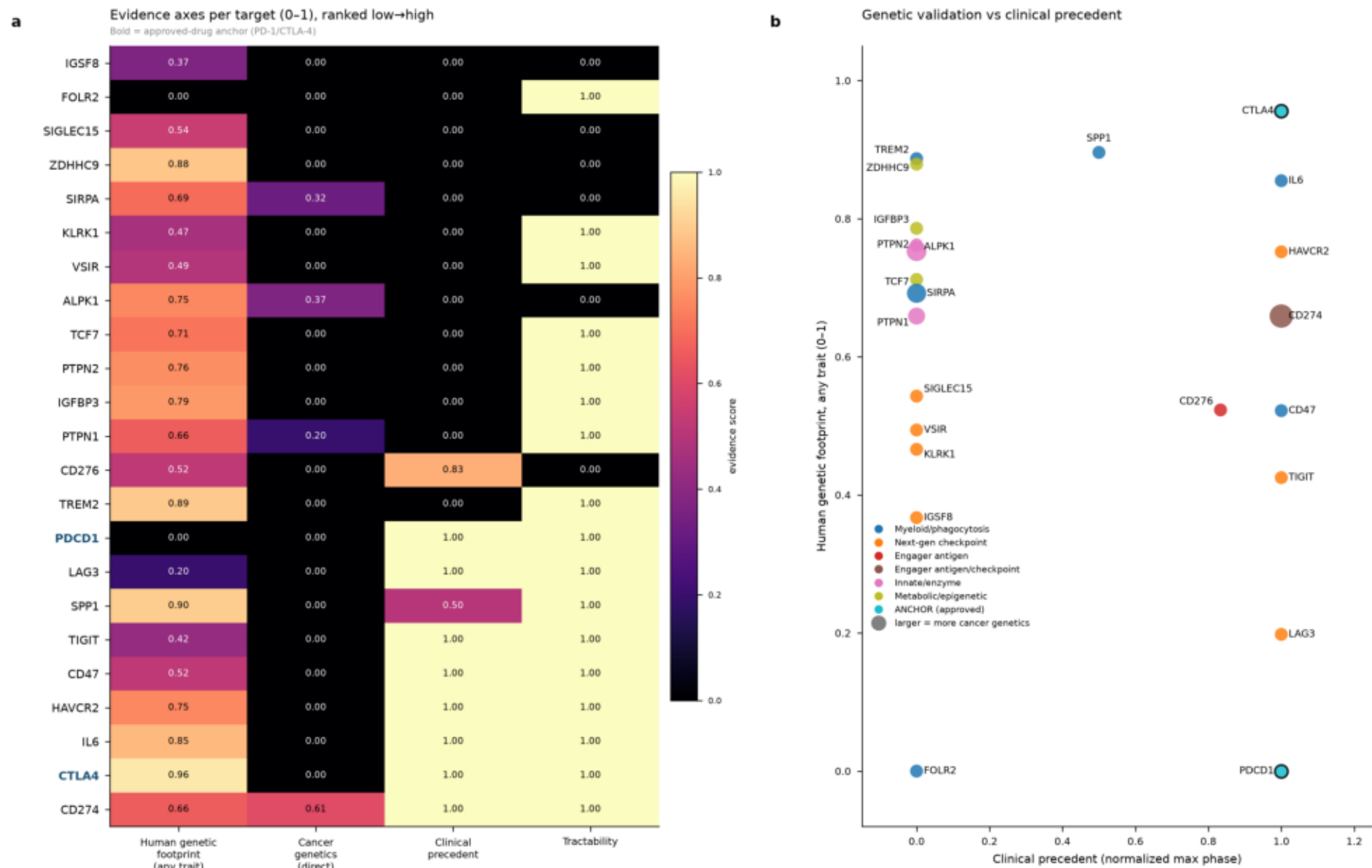
SPP1 (0.91), ZDHHC9 (0.89) and TREM2 (0.80) rank highest for a novel-target program; ZDHHC9 is genuine whitespace. No anti-osteopontin antibody is in oncology clinical development.

Bispecifics

Five spatially-rationalized designs exploit the SPP1+TAM ↔ exhausted-CD8 adjacency for avidity-driven niche selectivity. Lead: anti-CD44(SPP1-site) × anti-PD-1 (cis on exhausted CD8).

Genetic & functional-genomic evidence (Open Targets)

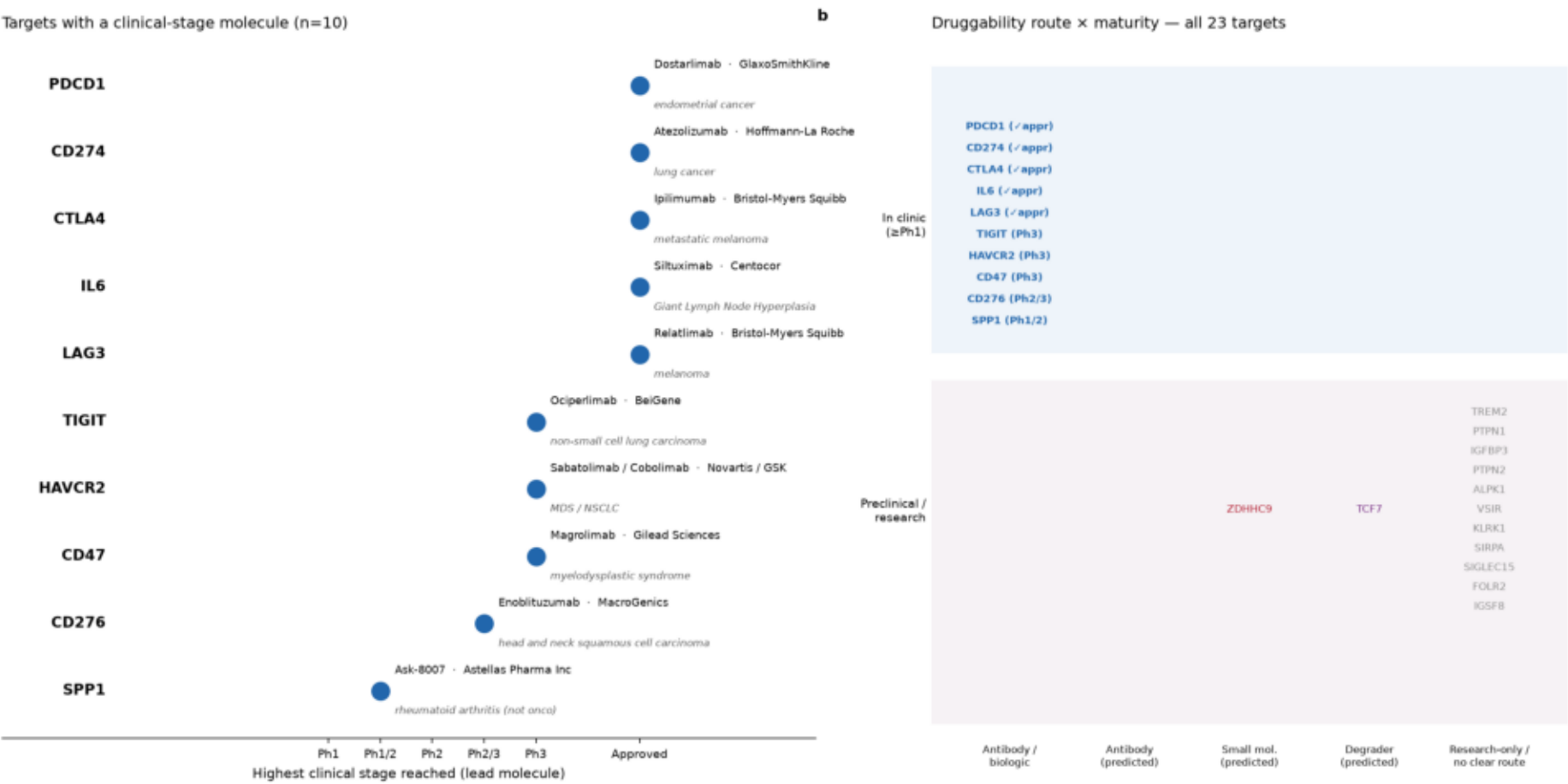
Genetic & functional-genomic evidence for recent onco-immunology targets (Open Targets)



Per-target evidence axes (0–1) and genetic validation vs clinical precedent. IO-modulator targets carry little direct germline/somatic cancer causation — even PD-1/CTLA-4 score 0 on cancer genetics.

Clinical feasibility — what is druggable, how, and how far

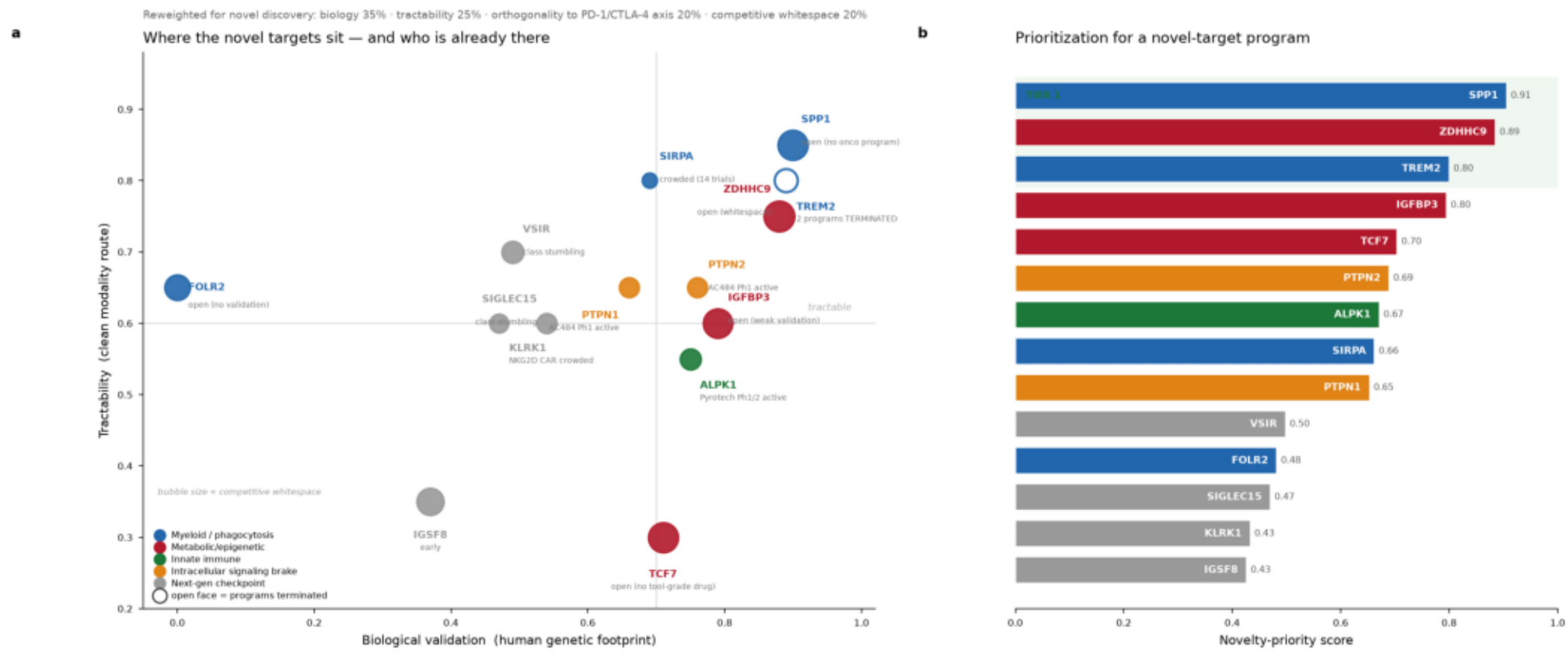
Clinical feasibility of the 23 onco-immunology targets: what is druggable, how, and how far it has advanced



10 of 23 targets have a clinical-stage molecule. SPP1's most-advanced agent (ASK-8007) reached Ph2 in rheumatoid arthritis — not oncology.

Strategy for the preclinical / novel target space

Strategy for the preclinical / novel onco-immunology target space

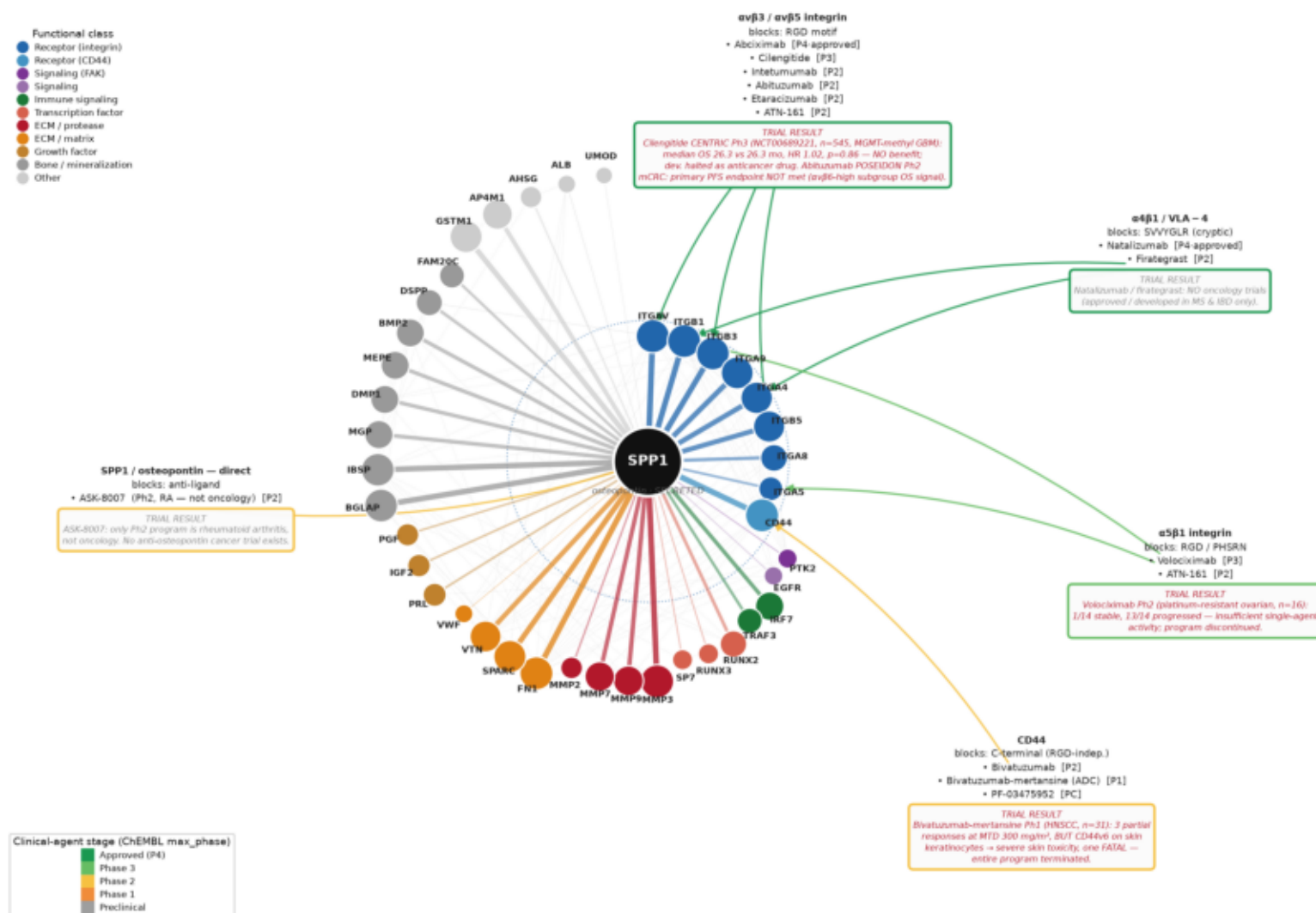


Rewighted for novel discovery (biology 35% · tractability 25% · orthogonality 20% · whitespace 20%). SPP1, ZDHHC9, TREM2 lead; open faces mark terminated programs.

SPP1 interaction network — druggable interfaces + trial outcomes

SPP1 (osteopontin) interaction network — druggable interfaces with clinical-trial outcomes

Every SPP1-receptor interface has been drugged, but no agent has succeeded on this axis in oncology: integrin trials were negative (cixigimab CENTRIC, volociximab, POSEIDON) and the CD44 ADC was fatally toxic.



boxed agents = real ChEMBL molecules on each SPP1-receptor interface (arrow → node blocked); TRIAL RESULT = documented oncology outcome (ClinicalTrials.gov + PubMed)

Every SPP1-receptor interface has been drugged, but none has succeeded in oncology: integrin trials negative, CD44 ADC fatally toxic, and no anti-osteopontin cancer trial exists.

Immune-metabolic pathways of three novel targets (Reactome / QuickGO)

Immune-metabolic pathways of three novel onco-immunology targets (Reactome / QuickGO curated)



SPP1 & ZDHHC9 branch from a single molecular node; TCF7 has two mechanistically distinct tracks — a WNT/ β -catenin-dependent transactivator (metabolic/proliferation) and a WNT-independent lineage-factor role (stem-like T-cell maintenance). Only the latter underlies its immunotherapy relevance.

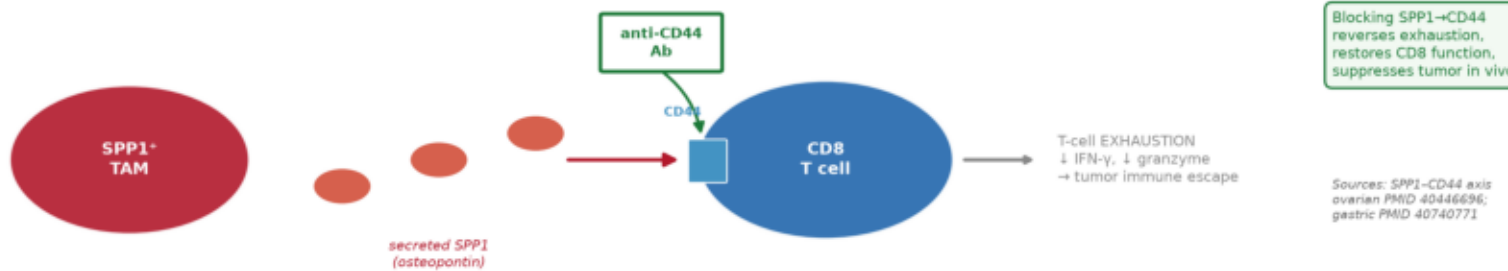
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Development plan — naked anti-CD44 blocking the SPP1-CD44 axis

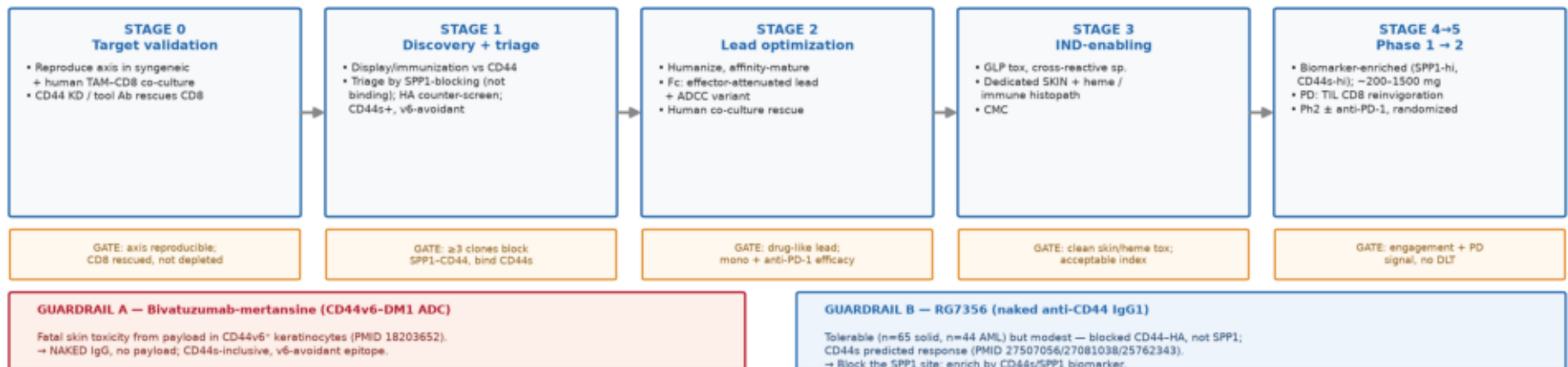
Development plan — a naked anti-CD44 antibody that blocks the SPP1-CD44 axis

Receptor-side alternative to ligand neutralization; engineered around two prior anti-CD44 clinical outcomes and the 2025 IO validation of SPP1-CD44

a Mechanistic rationale — the SPP1-CD44 immunosuppressive axis (IO-validated, 2025)



b Development cascade with go / no-go gates



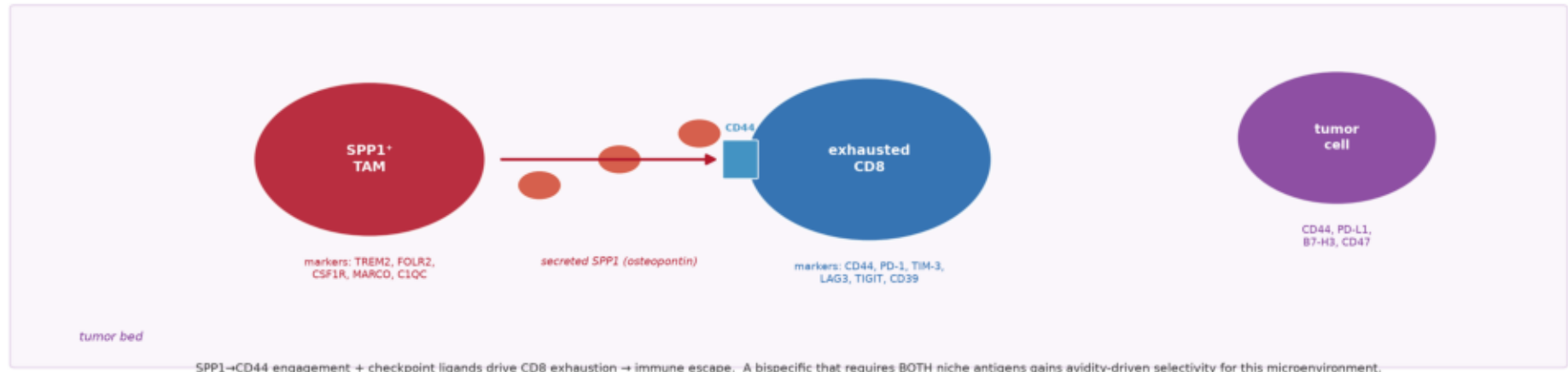
Mechanistic rationale + Stage 0-5 go/no-go cascade. Guardrail A (no payload — bivatuzumab lesson); Guardrail B (block SPP1 site not HA, biomarker-enrich — RG7356 lesson).

Spatially-rationalized bispecific designs

Spatially-rationalized bispecific antibody designs for the SPP1-CD44 immunosuppressive axis

Antigen compartments assigned from HPA single-cell + CELLxGENE CellGuide markers; niche adjacency from project spatial studies (gastric/liver-met/RCC/HGSOC)

a The immunosuppressive niche — SPP1⁺ TAM adjacent to exhausted CD8 (peer-reviewed spatial barrier)



b Five spatially-rationalized bispecific designs — arm A × arm B, with binding geometry

BsAb-1 • TIER 1 • lead-adjacent anti-CD44(SPP1-site) × anti-PD-1 geometry: cis on exhausted CD8 Two exhaustion inputs on ONE T cell. Both arms have safe-dosing precedent (RG7356; approved anti-PD-1). Avidity concentrates on PD-1⁺ exhausted CD8.	BsAb-2 • TIER 2 anti-SPP1(C-term) × anti-TREM2 geometry: cis on TAM — ligand sink TREM2 anchors Ab on the SPP1⁺ TAM; anti-SPP1 arm neutralizes osteopontin AT ITS SOURCE. Ligand arm = first-in-class for oncology (novelty + risk).	BsAb-3 • TIER 2 • novel geometry anti-CD44(SPP1-site) × anti-TREM2 geometry: trans — bridges CD8 ↔ TAM Dual engagement ONLY where exhausted CD8 abuts a TREM2⁺ SPP1⁺ TAM — i.e. inside the barrier. Built directly on the spatial adjacency (PMID 40740771).
BsAb-4 • TIER 2 anti-CD44(SPP1-site) × anti-TIM-3 geometry: cis on terminal-exhausted CD8 Targets the deepest-exhausted CD8 that PD-1 blockade misses; differentiated from crowded PD-1 space. Both arms clinically dosed (sabatolimab; RG7356).	BsAb-5 • TIER 3 anti-CD44(SPP1-site) × anti-CD39 geometry: cis on terminal-exhausted CD8/TAM Relieves SPP1→CD44 signal AND the adenosine brake on the same cells — two non-redundant suppressive pathways in one molecule. Anti-CD39 is clinical.	

Five arm-A × arm-B designs across three tiers. A bispecific requiring BOTH niche antigens gains avidity-driven selectivity for the immunosuppressive microenvironment.

Data sources & grounding

Every claim in this deck is grounded in a named database or peer-reviewed source:

- Drug / ligand data: ChEMBL.
- Clinical trials: ClinicalTrials.gov + PubMed.
- Genetic / functional-genomic evidence: Open Targets.
- Interaction networks: STRING.
- Cell-type expression / compartments: Human Protein Atlas + CELLxGENE CellGuide.
- Spatial adjacency (SPP1+ TAM ↔ exhausted CD8): peer-reviewed studies — gastric PMID 40740771, ovarian PMID 40446696, liver-met PMID 40335453, ccRCC PMID 40680256; anchor Klement JCI 2018 PMID 30395540.
- Patents: keyword-level web scan (NOT a professional FTO opinion).
- Caveat: spatial adjacency read from published figures/text, not recomputed from raw spots.